Sahaj Bhadja

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Summary

Solutions-oriented engineer experienced in **solution design, system architecture, requirements analysis, reusable components, and IT project implementation** from planning through testing and deployment in collaborative engineering teams.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 – Present*
- Translated business requirements into a multi-tenant platform design with normalized schemas and access controls for 500+ users, reducing dashboard latency from 1.2 seconds to 350 milliseconds through implementation-level tuning.
- Implemented cloud data pipelines processing 5,000+ monthly jobs with validation, reconciliation, and CI/CD-backed releases, increasing concurrent throughput 3x while achieving 99%+ reliability across planning, build, and deployment phases.

Grad8 **LA, California**
Software Engineer Intern *June 2025 – August 2025*
- Supported analysis, design, build, testing, and deployment of React and Flask product workflows, serving 150+ instructors and processing 10,000+ daily requests with stable production performance at scale.
- Evaluated solution alternatives for OCR, model, and API failures, implementing reusable modules, automated tests, and documented fallback paths while reliably reducing customer feedback turnaround time 60% at scale.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Contributed from requirements analysis through deployment for cross-platform Java and Python microservices, automating 20+ tests and Jenkins CI/CD while increasing pipeline efficiency 68% with Microsoft and GitHub stakeholders.
- Designed reusable object-oriented components using Singleton and Factory patterns, aligning API implementations with logical design while improving response times 30% and reducing technical debt 20% at scale.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Designed eight reusable microservices for authentication, resource allocation, billing, and notifications, synchronizing two million records through REST APIs, Azure SQL, Docker, and Jenkins-based staged deployments at scale.
- Collaborated across engineering and operations to automate testing and production releases, reducing deployment time 50% while using monitoring insights to reduce resource wastage 42% and increase conversions 15%.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*
- Analyzed client requirements, mapped 12+ integration points, and presented architecture recommendations before implementing cross-platform booking, payment, and real-time status capabilities aligned to the agreed logical design at scale.
- Delivered Flask and Firebase solutions across 50,000+ records, validating implementation against specifications while achieving sub-200-millisecond responses and reducing manual refund handling 70% for customers at production scale.

Skills

Solutions: Requirements analysis, solution design, system architecture, implementation design, reusable components, process improvement
Delivery: Planning, analysis, design, build, testing, deployment, CI/CD, Agile, documentation, stakeholder collaboration
Platforms: Python, Java, JavaScript, TypeScript, SQL, REST APIs, microservices, Docker, Kubernetes
Cloud/Data: AWS, Azure, GCP, PostgreSQL, SQL Server, Azure SQL, MongoDB, Jenkins, Git

Projects

Chat2SQL Code | *Platform Design, Reusable Components*
- Designed Java and TypeScript services supporting 10+ databases, configuring reusable Docker workflows across Windows, macOS, and Linux with schema synchronization, validation, and reliable cross-platform deployment at scale.

HMIS - Hospital Management Information System Live | *Operational Solution Design*
- Translated healthcare workflow requirements into TypeScript appointment, consent, and notification solutions, adding Sentry monitoring while reliably reducing record retrieval time 50% and administrative workload 30% for operational teams.

FinNexus Live | *Technology Platform Architecture*
- Designed a scalable TypeScript platform supporting 50,000+ users and one million transactions, implementing Plaid integrations, MongoDB persistence, Kubernetes deployment, and CI/CD automation for reliable delivery at scale.